

SiLU#

[https://pytorch.org/docs/stable/generated/torch.nn.modules.activation.SiLU.h](https://pytorch.org/docs/stable/generated/torch.nn.modules.activation.SiLU.html)

Description:

Applies the Sigmoid Linear Unit (SiLU) function, element-wise. The SiLU function is also known as the swish function. See Gaussian Error Linear Units (GELUs) where the SiLU (Sigmoid Linear Unit) was originally coined, and see Sigmoid-Weighted Linear Units for Neural Network Function Approximation in Reinforcement Learning and Swish: a Self-Gated Activation Function where the SiLU was experimented with later. Input: $(*)^{(*)}(*)$, where $*^{**}$ means any number of dimensions.

Function Signature

```
class torch.nn.modules.activation.SiLU(inplace=False)
[source]#
```

Shape:

```
extra_repr()[source]#
```

Return type

```
forward(input)[source]#
```

Returns:

Tensor

Code Examples

```
>>> m = nn.SiLU()  
>>> input = torch.randn(2)  
>>> output = m(input)
```

Notes & Warnings

NOTE See Gaussian Error Linear Units (GELUs) where the SiLU (Sigmoid Linear Unit) was originally coined, and see Sigmoid-Weighted Linear Units for Neural Network Function Approximation in Reinforcement Learning and Swish: a Self-Gated Activation Function where the SiLU was experimented with later.

Detailed Documentation

Documentation

Applies the Sigmoid Linear Unit (SiLU) function, element-wise.

The SiLU function is also known as the swish function.

See Gaussian Error Linear Units (GELUs) where the SiLU (Sigmoid Linear Unit) was originally coined, and see Sigmoid-Weighted Linear Units for Neural Network Function Approximation in Reinforcement Learning and Swish: a Self-Gated Activation Function where the SiLU was experimented with later.

Input: $(*)^{(*)} (*)$, where $*^{**}$ means any number of dimensions.

Output: $(*)^{(*)} (*)$, same shape as the input.

Return the extra representation of the module.

Runs the forward pass.